

Cluster Autoscaler — Easy KCNA Notes

Since you just learned **HPA** and **VPA**, this is the third important autoscaling concept.

1. What is Cluster Autoscaler?

Cluster Autoscaler (CA) automatically **adds or removes Nodes** in a Kubernetes cluster based on whether the cluster needs more or less capacity.

★ **Cluster Autoscaler = Automatically changes the number of Nodes.**

Simple picture

Low workload



2 Nodes



High workload



Not enough space for Pods



3 Nodes



2. Why do we need Cluster Autoscaler?

Imagine your Kubernetes cluster has:

Node 1

Node 2

And these Nodes are already full:

Node 1 → Full ❌

Node 2 → Full ❌

Now Kubernetes needs to run more Pods:

Pod 1

Pod 2

Pod 3

Pod 4

Pod 5

But there isn't enough capacity.

The Cluster Autoscaler can detect that Pods cannot be scheduled because of insufficient capacity and can request additional Nodes from the underlying infrastructure.

Pods need more space



Cluster Autoscaler



Add Node



Node 3 



Pending Pods can be scheduled

3. What exactly does Cluster Autoscaler scale?

This is the most important point:

HPA

 Pods

VPA

 CPU / Memory of Pods

Cluster Autoscaler

 Nodes

★ Easy memory:

HPA → Pods

VPA → Pod resources

Cluster Autoscaler → Nodes

4. Real-Life Example 🏠

Imagine a school.

Initially:

 Building 1

It has enough classrooms for the students.

Suddenly, many new students arrive:



The existing classrooms are full.

The school builds another building:

 Building 1

 Building 2

When the number of students becomes smaller, an unnecessary building can potentially be removed.

That's similar to **Cluster Autoscaler**.

5. Cluster Autoscaler + HPA ★★ ★

These two can work together.

Imagine:

Users increase



HPA



More Pods



Not enough Node capacity



Cluster Autoscaler



More Nodes

Complete example:

Initially:

Nodes:

 Node 1

 Node 2

Pods:



Traffic increases:

HPA



But there isn't enough room.

Node 1 → Full

Node 2 → Full

Some Pods → Pending

Cluster Autoscaler:

CA



Add Node 3



 Node 1

 Node 2

 Node 3

Now the pending Pods may be scheduled.

6. When Does Cluster Autoscaler Add Nodes?

A common situation is:

Pods cannot be scheduled because the existing Nodes don't have enough resources.

For example:

Pod needs:

CPU = 2 CPU

Available:

Node 1 = 0.5 CPU ❌

Node 2 = 0.5 CPU ❌

The Pod remains **Pending**.

Cluster Autoscaler can recognize that the cluster needs more capacity and increase the Node group size, if the environment supports it.

7. When Does It Remove Nodes?

Suppose you have:

 Node 1

 Node 2

 Node 3

 Node 4

But the workload becomes very small.

If Node 4 is no longer needed and its workloads can safely be moved elsewhere, Cluster Autoscaler may remove that Node.

Before:



After:



 Easy memory:

High demand → Add Nodes

Low demand → Remove unnecessary Nodes

8. Important: Cluster Autoscaler Does NOT Directly Create Pods

 Cluster Autoscaler is not responsible for creating additional application Pods.

For example:

HPA



Creates/requests more Pod replicas

Cluster Autoscaler:

Sees that Nodes don't have enough capacity



Adds Nodes

So their jobs are different.

9. Cluster Autoscaler Does NOT Increase CPU of a Pod

That's VPA's job.

VPA



Pod needs more CPU/Memory

Cluster Autoscaler:

CA



Cluster needs more Node capacity

10. Cluster Autoscaler and Cloud Providers ☁

In cloud environments, Nodes are often provided by a **Node group / autoscaling group**.

For example:

Kubernetes



Cluster Autoscaler



Cloud provider





New VM/Node

So Cluster Autoscaler can adjust the size of the underlying Node group according to its configuration and the cloud provider's capabilities.

11. Minimum and Maximum Nodes ★

Cluster Autoscaler commonly works with a configured range.

For example:

Minimum Nodes = 2

Maximum Nodes = 10

This means:

Minimum → 2 Nodes

Maximum → 10 Nodes

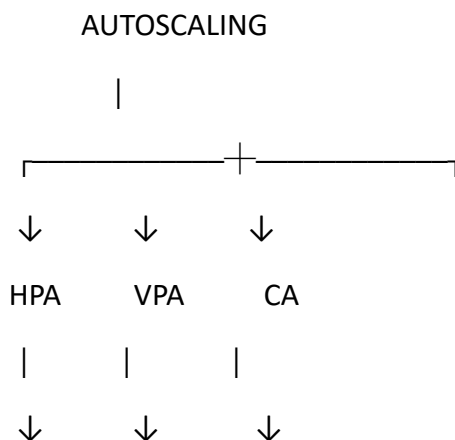
It won't scale below the configured minimum or above the configured maximum.

Example:

2 → 3 → 4 → 5 → ... → 10

12. HPA + VPA + Cluster Autoscaler

This is the most important comparison for your revision.



Pods CPU/Memory Nodes

HPA



More/Fewer Pods

VPA

Pod



More/Less CPU & Memory

Resources per Pod

Cluster Autoscaler



More/Fewer Nodes

★ KCNA Revision Notes

Cluster Autoscaler

- **CA = Cluster Autoscaler**
- Automatically adjusts the **number of Nodes** in a cluster.
- Adds Nodes when existing capacity is insufficient.
- Can remove Nodes that are no longer needed, when safe.
- Works with underlying infrastructure/cloud provider mechanisms.
- Has configured **minimum and maximum Node limits**.
- Often works together with **HPA**.

- HPA increases Pods → CA may add Nodes if there isn't enough capacity.
- CA does **not** directly scale the number of application replicas.
- CA does **not** directly increase CPU/memory assigned to a Pod.

Super Easy Memory

HPA →  Pods

VPA →  CPU / Memory

CA →  Nodes

★ **One-line exam definition:**

Cluster Autoscaler automatically adjusts the number of Nodes in a Kubernetes cluster based on the cluster's capacity requirements.